

A 28-GHz Low-Power Phased-Array Receiver Front-End With 360° RTPS Phase Shift Range

Robin Garg, *Member, IEEE*, and Arun S. Natarajan, *Member, IEEE*

Abstract—A low-power 28-GHz phased-array receiver (RX) front end is presented that incorporates a low-power low-noise amplifier (LNA) and a passive reflection-type phase shifter (RTPS) capable of 360° phase shift with 5-b phase resolution and low gain variation. Passive phase shifters are limited by tradeoffs between phase resolution, insertion loss, and phase shift range. The proposed RTPS load design and optimization approach leads to a 28-GHz RTPS achieving the state-of-the-art insertion loss with 360° phase shift range and low loss variation across the phase shift. The LNA adopts a transformer-coupled neutralization architecture that increases available gain, enabling lower power consumption. The phased-array front end is designed for *Ka*-band applications and has been implemented in 65-nm CMOS. The measured RTPS achieves 360° phase shift with 7.75 ± 0.3 dB insertion loss and an rms phase error of 0.3° at 28 GHz. The low-power phased-array RX front end has an overall gain of 9.5 ± 0.4 dB and noise figure < 5.5 dB at 28 GHz. The RX front end consumes 10 mW from a 0.9-V supply with phase shifter and an LNA active area of 0.16 and 0.32 mm², respectively, in 65-nm CMOS, demonstrating its suitability for low-power phased-array RX for emerging wireless links.

Index Terms—Beamforming, CMOS, complete 360° phase shifter, 5G cellular communications, gate-drain capacitance cancellation, *Ka*-band, low insertion loss and rms phase error, low power, low-noise amplifier (LNA), millimeter-wave reflection-type phase shifter (RTPS), neutralization LNA, passive phase shifter, phased-array receiver (RX), π -load, scalable arrays, switched transmission-line phase shifter.

I. INTRODUCTION

PHASED arrays enable electronically steerable beamforming, thereby providing higher effective isotropic radiated power (EIRP) in transmitters (TXs) and improved receiver (RX) signal-to-noise ratio (SNR) [1], [2]. The availability of large spectrum at millimeter-wave (mm-wave) has led to research on integrated mm-wave TX and RX. However, high mm-wave propagation loss in addition to limitations in TX output power and RX noise figure (NF) in integrated mm-wave TX/RX imply that the higher EIRP and improved

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R. Garg was with the School of Electrical Engineering and Computer Science, Oregon State University, Corvallis, OR 97330 USA. He is now with Cypress Semiconductor Corporation, Lynnwood, WA 98087 USA (e-mail: gargro@oregonstate.edu).

A. S. Natarajan is with the School of Electrical Engineering and Computer Science, Oregon State University, Corvallis, OR 97330 USA (e-mail: nataraja@eecs.oregonstate.edu).

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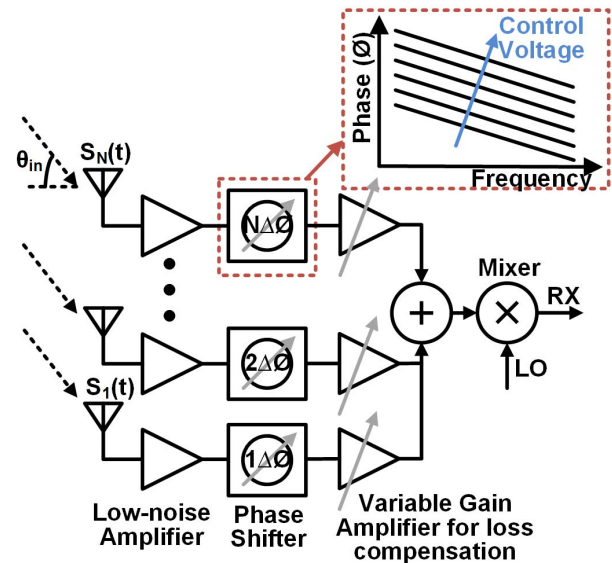


Fig. 1. RF-path phase-shifting architecture for phased-array RX.

SNR of arrays is often required to close wireless link budgets. The physically small antenna size and spacing makes mm-wave arrays feasible for consumer communication and radar applications. Initial array demonstrations at 24 and 77 GHz [1], [3]–[5] have led to large, scalable arrays at mm-wave frequencies such as 60 GHz [6] and 94 GHz [7], [8]. In recent years, there has been increasing interest in mm-wave networks for the next-generation wireless networks (5G) beyond 4G LTE with system-level measurements and demonstrations at 28 GHz [4], [9], [10]. Such networks require arrays for robust high-data rate mobile communications. In particular, mobile and large-scale array applications [11] require the array RX, as shown in Fig. 1, to simultaneously achieve low power consumption with low NF and phase shift capabilities. This has led to interest in RF-path phase-shifting implementations that promise low area and power consumption.

A critical building block in RX arrays with RF-path phase shifting is the RF variable phase shifter (VPS) that must provide 360° phase shift range in addition to low insertion loss, high linearity, and low power consumption [12]. Typical VPS approaches provide discrete phase shift steps with gain variation across phase shift settings. For most communication and radar applications, a phase shift resolution of 5 b (11.25°) in the phase shifter is considered sufficient to ensure negligible main-lobe array gain degradation (< 0.1 dB) across all scanning angles [13]. Variable gain across phase shift settings leads

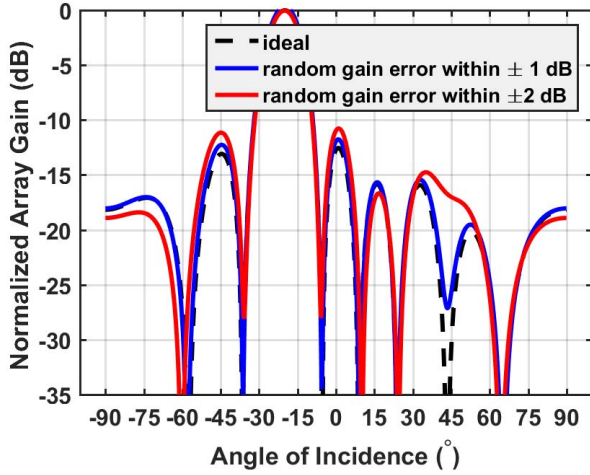


Fig. 2. Impact of gain variation across array elements on sidelobe and nulls in array factor in an eight-element linear array (interelement spacing: $\lambda/2$).

to variable gain across array elements when element phase shift settings are different. The impact of such variation on array gain is shown in Fig. 2, where element gain variation is modeled by assuming random gain variation in each element within a maximum gain error. As shown in Fig. 2, for an eight-element array, increased gain variation leads to increased sidelobe levels and poorer peak-to-null ratios in the array.

The state-of-the-art mm-wave VPS approaches include vector-modulator-based active [8], [14]–[16] and reflection-type phase shifter (RTPS) [17]/switched-transmission line phase shifter (STPS)-based [18] passive phase shifters. The inherent linearity and the zero power consumption of passive VPS make them attractive. However, passive VPS must ensure: 1) small area; 2) uniform loss across phase shift to prevent beam pattern degradation and/or eliminating the need for a variable-gain block preceding or succeeding it; and 3) high phase shift resolution to ensure accurate beamforming [12].

In this paper, we present a systematic approach to optimize the performance of a passive RTPS targeting full 360° phase shift with 5-b phase resolution at 28 GHz. With this approach, we achieve the state-of-the-art 7.75 ± 0.3 dB insertion loss in a commercial 65-nm CMOS process at 28 GHz while achieving 5-bit phase resolution, demonstrating the feasibility of our approach for *Ka*-band RX arrays. The RTPS is incorporated into a single-channel ~ 10 -mW RX-front-end targeting 5G applications. The front end includes a low-power three-stage low-noise amplifier (LNA) and achieves 9.5-dB gain and < 5.5 dB NF at 28 GHz.

Section II details passive VPS design challenges and places this paper in the context of the state of the art. The proposed approach to overcome nonuniform phase-shifter loss and achieve 360° phase shift range with low loss is described in Section III. A specific RTPS implementation at 28 GHz in 65-nm CMOS using this approach is described in Section IV. Section V describes a low-power 28-GHz RX front end incorporating the RTPS. Measured performance for the RTPS and the RX array front end is presented in Section VI. Finally, conclusion and future avenues of research are highlighted in Section VII.

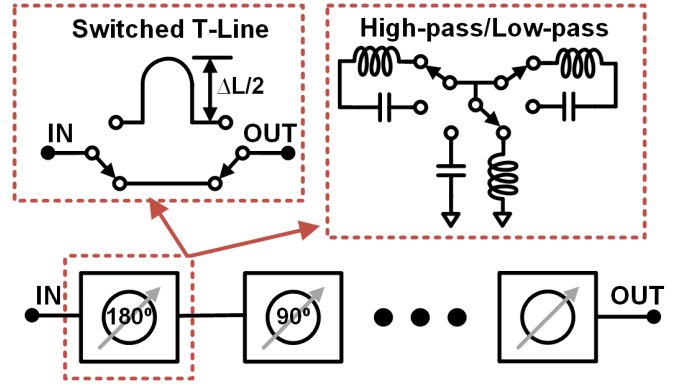


Fig. 3. Switch-type VPS using switched T-lines or high-pass/low-pass signal paths.

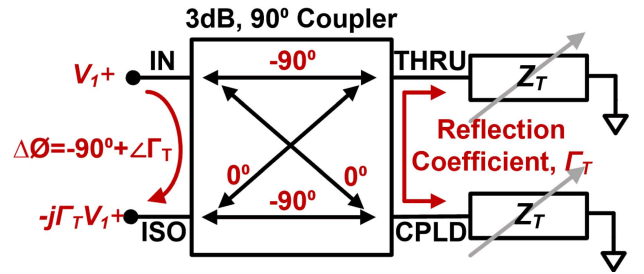


Fig. 4. RTPS using 3-dB quadrature coupler.

II. PASSIVE PHASE SHIFTERS FOR RF/mm-WAVE BEAMFORMING

Recent integrated passive phase shifters can be broadly divided into STPS (Fig. 3) and RTPS (Fig. 4) approaches. The improving f_T and switch time constant for CMOS transistors has translated to low-loss STPS (6.1–7.6 dB at 60 GHz [19], 5.6–7.6 dB at 28 GHz with 4-b resolution [20]). However, STPS phase resolution is limited by the number of switched signal paths (T-line or high-pass/low-pass), as shown in Fig. 3. Therefore, achieving higher resolution implies higher path loss and larger area [21]. In addition, varying loss across STPS switch states translates to loss variation across phase shift settings. For example, the 4-b STPS in [20] achieves 2-dB loss variation with $\sim 9^\circ$ rms phase error.

Integrated RTPS achieve targeted phase shift by varying through and coupled-port termination impedance for a quadrature 3-dB coupler [22]. As shown in Fig. 4, varying the reflection coefficient of the coupled-port and through-port loads leads to varying phase shift between the input and the output (input and isolated ports, respectively, of the hybrid coupler). In integrated implementations, the variable termination impedance is often achieved by changing control voltage across a varactor. Therefore, high resolution can be achieved without any area or loss penalty, particularly with on-chip implementations that allow CMOS digital-to-analog converter (DAC) with very high-voltage resolution.

However, varactor quality factor, Q , and loss vary across capacitance settings leading to phase shift dependent loss in the RTPS [23], [24]. As suggested in [7], loss variation across

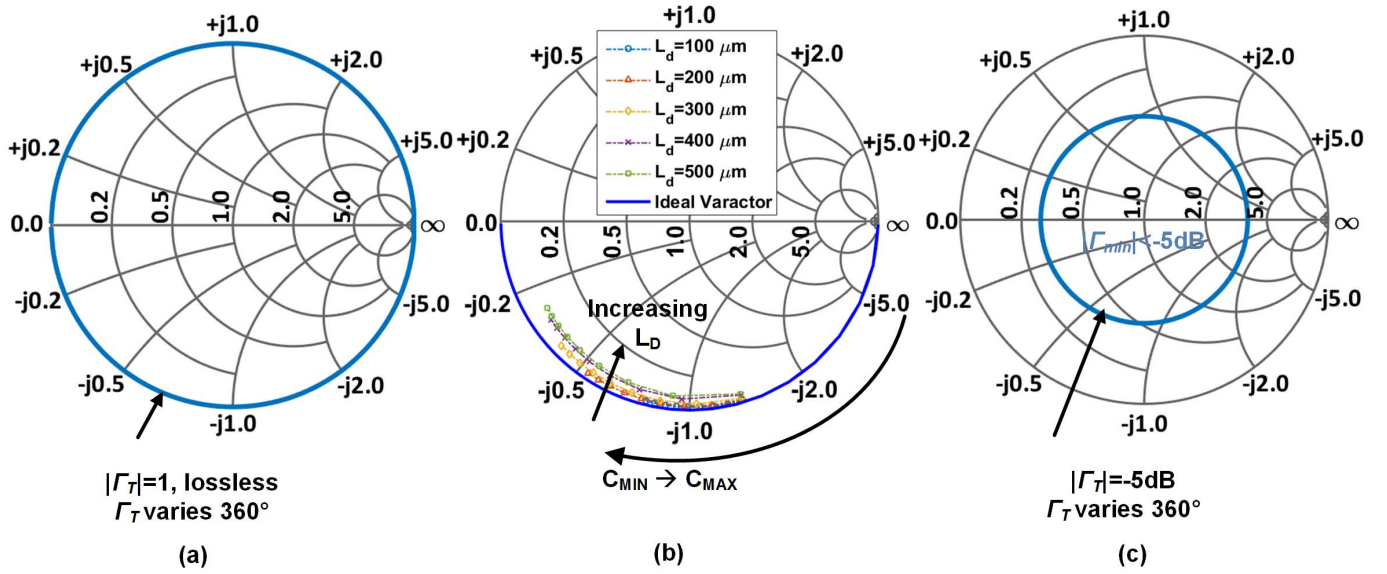


Fig. 5. Variable load represented on a Smith chart. (a) Lossless load corresponding to $|\Gamma_T| = 0$ dB with 360° phase shift range. (b) Limited phase shift range available when a single varactor is used to provide variable termination. (c) Load with loss corresponding to $|\Gamma_T| = -5$ dB and 360° phase shift range.

phase shift can be reduced by independently controlling two varactors in a π -network variable load. However, achieving sufficient varactor variation to ensure 360° phase shift is challenging. This results in the state-of-the-art on-chip implementations targeting a smaller phase shift range of $\sim 180^\circ$ while adopting higher order loads, such as a shunt- LC [24] and π -networks [17], [25]. The RTPS [26], [27] can then be combined with active phase shifters or STPS [7], [28], [29] in order to achieve full 360° phase shift. However, if the RTPS is followed by an active $0^\circ/180^\circ$ stages in such implementations [7], [23], the resultant nonlinearity/power consumption reduces some of the advantages of using a passive VPS. In this paper, we extend the approach in [7] by developing a systematic design methodology that reduces loss and ensures 360° phase shift.

III. PROPOSED REFLECTION-TYPE PHASE-SHIFTER ARCHITECTURE

Our objective is to achieve 360° phase shift range at 28 GHz while minimizing insertion loss. Assuming an ideal hybrid coupler, the phase shift, $\Delta\Phi$, and insertion loss (in dB), IL , through an RTPS (Fig. 4) are provided by

$$\Delta\Phi = -90^\circ + \angle\Gamma_T \quad (1)$$

$$IL = -20 \log(|\Gamma_T|) \quad (2)$$

where Γ_T is the reflection coefficient at the coupled and through ports. Therefore, $|\Gamma_T| = 1$ and $\angle\Gamma_T$ range of 360° [shown in Fig. 5(a)] is required for lossless operation across entire phase shift range. However, in practice, performance is limited by coupler insertion loss, finite isolation between input and isolated ports, and loss/phase variation limits in the variable reflective load. A single varactor-based reflective load requires a $C_{\text{max}}/C_{\text{min}}$ of ∞ in order to provide 180° variable phase, with the load moving from an open to a short on the

Smith chart, as shown in Fig. 5(b) [7]. Given resistive and capacitive parasitics, varactors with channel length varying from 100 to 500 nm achieve $C_{\text{max}}/C_{\text{min}}$ of 1.7–4.8 and a quality factor of 24 to 6 in CMOS technologies [30]. As shown in Fig. 5(b), this limits phase shift range to much less than 180° . Therefore, a higher order load is required to achieve large phase shift variations. However, based on (2), achieving a targeted insertion loss requires $|\Gamma_T| > |\Gamma_{\text{MIN}}|$ for all phase shift settings, as shown in Fig. 5(c).

A variable termination consisting of shunt inductor in parallel with varactor has been proposed to achieve increased phase shift range [24]. However, phase shift range is still $\sim 180^\circ$ with larger phase shift range leading to high insertion loss. Larger phase shift range is possible using the pi-network shown in Fig. 6(a), where the series combination of L_1 and C_2 can be viewed as a variable reactance in parallel with C_1 . This network can be viewed as an extension of [7] and [24]. In the following, we develop RTPS design constraints to minimize insertion loss for $\sim 360^\circ$ phase shift range assuming the π -network reflective load in Fig. 6(a). A similar constraint-based approach can also be adopted for other reflective loads topologies to minimize insertion loss across phase shift.

A. Impact of Varactor and Inductor Loss

Assuming a lossless coupler, losses in the varactor and inductor in Fig. 6(a) lead to $|\Gamma_T| < 1$. Assuming a termination impedance, Z_T (Fig. 4), and a characteristic impedance, Z_0 , the reflection coefficient, Γ_T , is given by

$$\Gamma_T = \frac{Z_T - Z_0}{Z_T + Z_0} \quad (3)$$

From (3), assuming $Z_T = R_T + jX_T$, it can be shown that to ensure $|\Gamma_T| > |\Gamma_{\text{MIN}}|$

$$(R_T - A)^2 + X_T^2 > R^2 \quad (4)$$

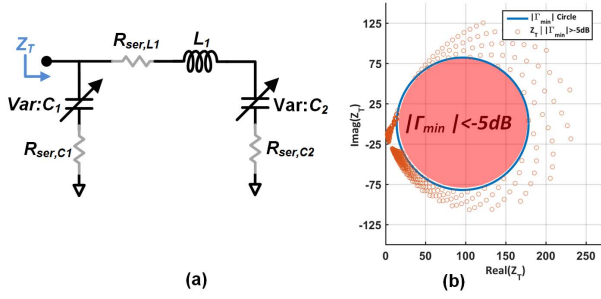


Fig. 6. (a) Variable π -network load based on varactors. (b) Impedance circle corresponding to $|\Gamma_T| = -5$ dB based on (3).

where

$$A = Z_0 \frac{1 + |\Gamma_{\text{MIN}}|^2}{1 - |\Gamma_{\text{MIN}}|^2} \quad (5)$$

$$R = 2Z_0 \frac{|\Gamma_{\text{MIN}}|}{1 - |\Gamma_{\text{MIN}}|^2}. \quad (6)$$

Therefore, the Γ_T constraint in Fig. 5(c) translates graphically to Fig. 6(b), where acceptable values of Z_T that ensure $\Gamma_T > \Gamma_{\text{MIN}}$ lie outside the circle defined by (4). For the network in Fig. 6(a), Z_T is given by

$$Z_T = \frac{1 + s^2 L_1 C_2}{s(C_1 + C_2) + s^3 L_1 C_1 C_2}. \quad (7)$$

From (3) and (7), $\angle \Gamma_T$ is given by

$$\angle \Gamma_T = 2 \tan^{-1} \left(\frac{\omega Z_0 (C_1 + C_2) \left(\omega^2 L_1 \frac{C_1 C_2}{C_1 + C_2} - 1 \right)}{1 - \omega^2 L_1 C_2} \right). \quad (8)$$

As noted in [7], C_1 and C_2 in Fig. 6(a) can be varied independently to minimize loss across phase shift. Additionally, coupler impedance Z_0 is also a design variable impacting RTPS phase shift, as shown in (8). Therefore, Z_0 can be potentially selected to optimize phase shift range and insertion loss. However, in this paper, Z_0 is set to 50Ω to simplify mm-wave test. In Section IV, design constraints are developed to select design parameters, L_1 , C_1 , and C_2 at $\omega = 28$ GHz. While parameter selection according to the proposed approach and detailed simulations including component parasitics are discussed in Section IV, we utilize parameter values close to the final design values in Section IV in this section to illustrate the proposed approach. The simplified network model in Fig. 6(a) includes resistors to include component loss. In the following, we assume $L_1 = 425$ pH ($R_s = 5.3$ for Q of 14), $C_1 = 80$ fF $\rightarrow$ 264 fF, and $C_2 = 45$ fF $\rightarrow$ 149 fF. The series resistance of 2 and 3.6Ω is added to varactors C_1 and C_2 , respectively ($Q \sim 10$) to model varactor losses.

Fig. 6(b) shows Z_T for the network in Fig. 6(a) at 28 GHz using (7) and component parameters detailed earlier. While inductor and varactor losses degrade $|\Gamma_T|$, selecting C_1 and C_2 such that Z_T lies outside the circle in Fig. 6(b) can ensure that all phase shift settings are achieved with ~ 5 dB insertion loss. This can also be represented graphically in terms of the insertion loss and phase shift across C_1 and C_2 settings, as shown in Fig. 7. Notably, the targeted uniform loss is

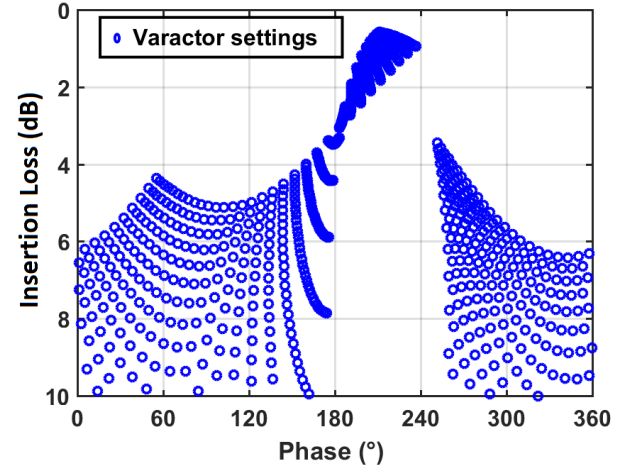


Fig. 7. RTPS gain and phase shift across varactor voltage for π -network load in Fig. 6(a).

limited by the *worst case loss* across phase shift, since C_1 and C_2 values for all phase shifts are selected to achieve this loss to ensure low loss variation. Such an approach is also followed in the design of other phase shifter topologies. For example, STPS equalizes loss for ON and OFF settings, or active phase shifters equalize loss by discarding high gain settings. Therefore, minimizing insertion loss requires that the worst-case loss must be minimized.

B. Impact of Isolation in Coupler

In addition to losses associated with the variable terminations, an RTPS with integrated hybrid coupler also suffers from insertion loss due to coupler dielectric/metal losses. Additionally, finite isolation between input and isolated port presents a leakage path between RTPS input and output in addition to the reflected signals from through and coupled ports. Since the relative phase between the leakage signal and reflected signal depends upon Γ_T , the finite isolation leads to a phase-shift-dependent loss through the RTPS.

In the following, S -parameters derived from electromagnetic (EM) simulations of the implemented lumped-element quadrature hybrid coupler in Section IV are used to demonstrate the impact of coupler loss (~ 1 dB) and finite coupler isolation (~ 17.7 dB in simulation). Fig. 8 shows the RTPS loss with ideal lossless terminations and EM-simulation derived coupler model. The RTPS loss, which is due to metal loss and finite isolation in the coupler, varies across phase shift settings with a minimum due to destructive addition of leakage and reflected signals as discussed earlier. Fig. 8 also plots the lowest RTPS insertion loss across phase shift if an ideal coupler and lossy termination are assumed (obtained from Fig. 7). Worst-case RTPS insertion loss can be improved by aligning the phase shift setting corresponding to the lowest termination loss with the phase shift setting corresponding to the highest insertion loss due to the coupler, as shown in Fig. 8. For example, since the loss due to coupler is highest for $\Phi = 180^\circ$ for the coupler implemented in this paper, L_1 , C_1 , and C_2 must be selected to achieve lowest insertion loss at $\Phi = 180^\circ$. Fig. 9 shows the simulated insertion loss across

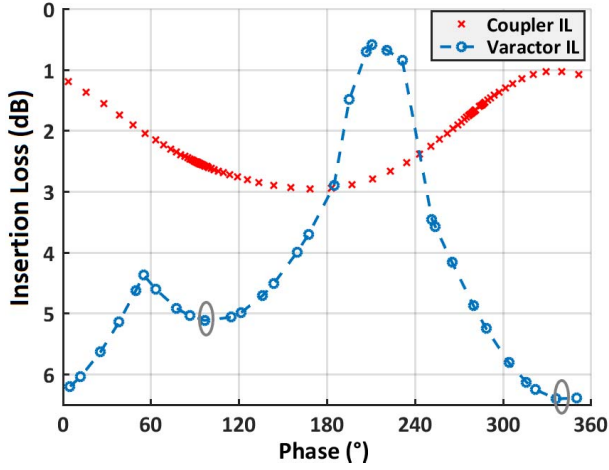


Fig. 8. RTPS insertion loss due to coupler losses/isolation and best case insertion loss due to π -network load across RTPS phase shift from Fig. 7.

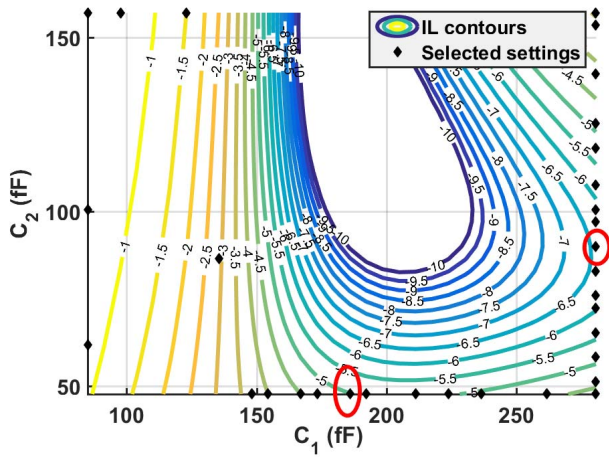


Fig. 9. Insertion loss contours across C_1 and C_2 —the selected points for the best case loss across phase shift shown in Fig. 8 are also plotted. Contours indicated lowest loss when C_1 is low and C_2 is high.

C_1 and C_2 for component values described in Section III-A illustrating that minimum loss in Fig. 8 occurs when C_1 is at its minimum and C_2 is at its maximum. Our objective is to ensure that this corresponds to the phase shift with the highest coupler insertion loss, which imposes the following constraint.

Constraint 1: The value of L_1 and the varactors C_1 and C_2 must be selected such when $C_1 = C_{1,\min}$ and $C_2 = C_{2,\max}$, the phase shift through the RTPS, $\Delta\Phi \sim 180^\circ$.

Additionally, as shown in Fig. 8, insertion loss due to varactor losses shows two local minima. It is also desirable to balance the insertion loss in these two cases to improve worst case performance. For instance, for the set of values circled in red in Fig. 9, these local minima occur at ($C_1 = C_{1,\text{mid}}$ and $C_2 = C_{2,\min}$) and ($C_1 = C_{1,\max}$ and $C_2 = C_{2,\text{mid}}$). Therefore, an additional design constraint can be targeted.

Constraint 2: The range of C_1 and C_2 should be selected, such that the two minima in the varactor loss curve in Fig. 8 are balanced to minimize worst case loss.

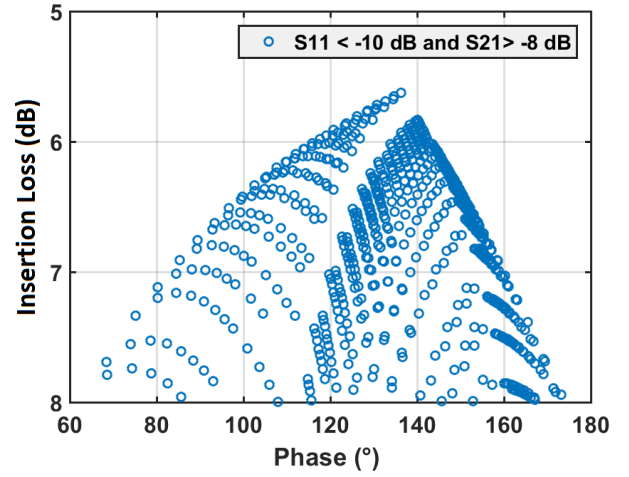


Fig. 10. Simulated RTPS insertion loss and phase shift across varactor voltage for unequal terminations at thru and coupled ports while ensuring better than 10-dB return loss at input and output.

C. Phase Shift Variation Requirements

The constraints associated with insertion loss must be satisfied in the context of the targeted phase shift range. Phase shift range is closely related to the varactor $C_{\max}/C_{\min}$. Increasing MOS varactor channel length results in larger $C_{\max}/C_{\min}$ at the cost of poorer varactor Q [30]. In the following, we assume a worst case varactor Q of 10. Based on stand-alone varactor simulations and [30], a channel length of 300 nm is selected with simulated $Q \sim 12$, providing design margin for wiring losses and model errors. This channel length corresponds to a $C_{\max}/C_{\min}$ ratio of 3.3, which is used in the following analysis.

Defining the ratio of the varactors in the minimum setting as

$$k = C_{\min,1}/C_{\min,2} \quad (9)$$

and assuming $C_{\max}/C_{\min} = 3.3$ at 28 GHz, the phase shift, $\angle\Gamma_{T,1}$, for $C_1 = C_{\min,1}$ and $C_2 = C_{\min,2}$, can be calculated from (8) and (9) as

$$\angle\Gamma_{T,1} = 2 \tan^{-1} \left(\frac{\omega Z_0 C_{\min,1} (k+1) \left(p^{\frac{k}{k+1}} - 1 \right)}{1 - p \cdot k} \right) \quad (10)$$

where

$$p = \omega^2 L_1 C_{\min,1}. \quad (11)$$

Correspondingly, the phase shift at the maximum capacitance setting ($C_1 = C_{\max,1}$ and $C_2 = C_{\max,2}$) is provided by

$$\angle\Gamma_{T,2} = 2 \tan^{-1} \left(\frac{\omega Z_0 3.3 C_{\min,1} (k+1) \left(p^{\frac{3.3k}{k+1}} - 1 \right)}{1 - 3.3 p \cdot k} \right). \quad (12)$$

The contours of phase shift range ($\angle\Gamma_{T,1} - \angle\Gamma_{T,2}$) across k and p for different values of $C_{\min,1}$ are shown in Fig. 11(a) and (b) based on (10) and (12). Notably, only certain combinations of k and p achieve targeted phase shift for given $C_{\min,1}$. Therefore, the following constraint can be targeted.

Constraint 3: The choice of k and p must ensure targeted phase shift range at frequency of interest.

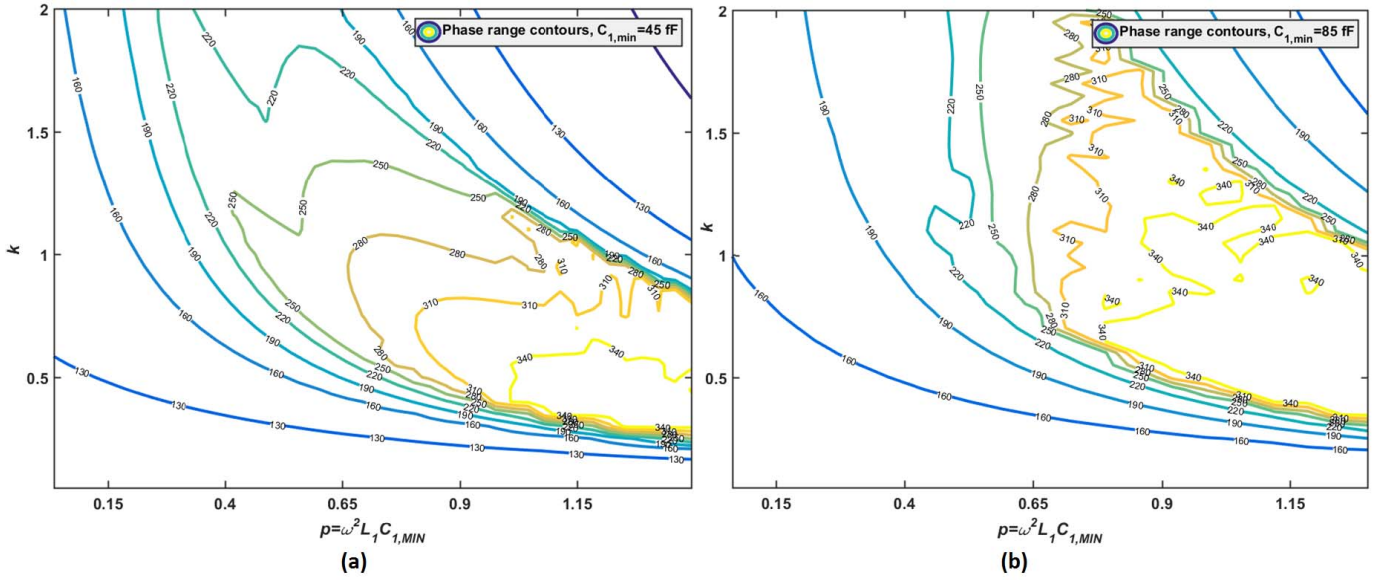


Fig. 11. Phase shift range contours using simplified π -network termination model across k and p for (a) $C_{1,\min} = 45$ fF and (b) $C_{1,\min} = 85$ fF.

D. Unequal Terminations for Increased Phase Shift Range

In addition to the above constraints, independent control of capacitors in through- and coupled-port termination can be leveraged to provide unequal terminations at the through and coupled port. This approach can also lead to variable phase shift across termination impedances. However, unequal terminations at through and coupled ports lead to poorer RTPS matching, which also increases insertion loss. Fig. 10 shows simulated RTPS insertion loss and mismatch with unequal terminations while limiting RTPS input and output return loss to 10 dB. This constraint results in a smaller phase shift range by limiting the acceptable range of unequal terminations. However, this approach can be used to provide additional phase shift range to overcome modeling errors and impact of layout parasitics.

IV. 28-GHz RTPS DESIGN IN 65-nm CMOS

In Section V, we discuss the design of a 28-GHz RTPS in the context of the constraints described earlier to minimize insertion loss across phase shift range. The three design variables, $C_{1,\min}$, $C_{2,\min}$, and L_1 , are transformed to the variables p , k and $C_{1,\min}$ based on (9) and (11). As mentioned in Section III-C, a varactor capacitance ratio of 3.3 is assumed.

Since the phase shift for given C_1 , C_2 , and L_1 is given by (8), satisfying constraint 1 in Section III-B implies

$$1 - \omega^2 L_1 C_{2,\max} = 1 - 3.3p \cdot k = 0. \quad (13)$$

While (13) relates k and p , the parameters must also satisfy constraint 3. Fig. 11 shows contour plots for phase shift range across k and p for two different values of $C_{1,\min}$ (45 and 85 fF). As shown in Fig. 11, phase shift range is maximized for $0.3 \leq k \leq 0.6$ for $C_{1,\min} = 45$ fF and $0.4 \leq k \leq 1.0$ for $C_{1,\min} = 85$ fF. From (13), which relates k and p , the targeted range for k translates to $0.5 \leq p \leq 1.0$ and $0.3 \leq p \leq 0.75$,

respectively. This represents a conflict, since the range of p to achieve 360° phase shift range from Fig. 11 is larger. This conflict can be addressed partially by using Fig. 11 to select a smaller value for k for a given value of $C_{\min,1}$. This enables a higher value for p based on (13), thereby increasing overall phase shift range.

Implementation constraints limiting practical inductance values also affect the choice of C_1 . Given $C_{1,\min}$ and p , L_1 can be calculated from (11). For example, for $C_{1,\min} = 45$ fF, achieving $p > 0.7$ translates to $L_1 \geq 500$ pH at 28 GHz. Integrating such a large inductance at mm-wave is challenging due to the parasitic capacitances and resistance of the inductor metal traces. Increasing $C_{1,\min}$ decreases the required L_1 value to achieve targeted p . For example, based on Fig. 11, $C_{\min,1} = 85$ fF is preferable for lower L_1 . Notably, balancing the minima for Constraint 2 in Section III-B also dictates the choice of $C_{1,\min}$.

In this paper, we initially assume a $C_{1,\min}$ range and compute L_1 and C_2 based on $k \sim 0.55$ and $p \sim 0.9$. Based on insertion loss plots across C_1 and C_2 settings, we select $C_{1,\min} = 85$ fF, and $k = 0.56$, $p = 1.11$ leading to $L_1 = 425$ pH and $C_{2,\min} = 45$ fF.

Fig. 12 shows the schematic of the 28-GHz RTPS implementation in the 65-nm CMOS technology. The transformer-based lumped-element quadrature hybrid coupler is implemented based on the design approach described in [31] (Fig. 13). The even-mode and odd-mode impedances of 217.3 and 16.1 Ω , respectively, are targeted for $Z_0 = 50 \Omega$. EM simulations using HyperLynx IE3D EM solver were performed to model the transformer, which was implemented using the top metal layer of thickness 3.4 μm with crossovers in a 0.9- μm -thick metal layer [Fig. 13(a)]. Given the 28-GHz operating frequency, wiring parasitics are modeled as parasitic inductances and capacitances. Capacitances C_p and C_m are implemented using MIM capacitors with the density

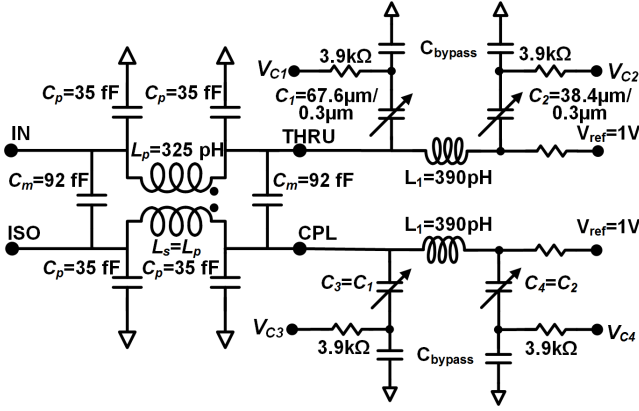
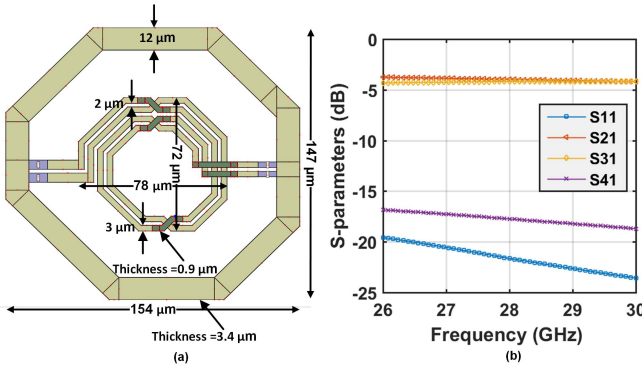


Fig. 12. Schematic of 28-GHz RTPS implementation in 65-nm CMOS.

Fig. 13. Layout of transformer in coupler shown in Fig. 12. (a) Transformer is implemented in the top metal layer (thickness: 3.4 μm). (b) Simulated coupler performance using EM model of transformer along with MIM capacitors shown in Fig. 12.

of $\sim 2 \text{ fF}/\mu\text{m}^2$ that are available as part of the design kit. C_p occupies $4 \mu\text{m} \times 4 \mu\text{m}$, while C_m occupies $9.2 \mu\text{m} \times 4.6 \mu\text{m}$. Simulated coupler parameters based on EM model for the transformer are shown in Fig. 8. While stand-alone coupler test structures were not fabricated, measured coupler performance is captured by the comparison between overall measured and simulated RTPS performance in Fig. 19.

A two-turn inductor occupying $146 \mu\text{m} \times 163 \mu\text{m}$ is used to implement L_1 . In order to accommodate interconnect parasitics, L_1 is lowered to 390 pH with remaining inductance coming from intercomponent wiring. Varactors, C_1 and C_2 , are thin-oxide mos varactors implemented using the length of $0.3 \mu\text{m}$ and the finger width of $1.6 \mu\text{m}$ (C_1 : 42 fingers $\times$ $1.6 \mu\text{m}$ and C_2 : 24 fingers $\times$ $1.6 \mu\text{m}$).

Symmetric variation of C_1 and C_2 in the through-port and coupled-port terminations is accomplished by ensuring $V_{C1} = V_{C3}$ and $V_{C2} = V_{C4}$. As can be seen from Fig. 11(b), the simulated phase shift range with such choices of k and p only achieves $\sim 330^\circ$ phase shift range, falling short of the targeted 360° . Hence, symmetric variation of the coupled-port and through-port terminations leads to 4-b phase resolution. Therefore, we propose to increase the phase shift range from $\sim 330^\circ$ to 360° by using unequal terminations as described in Section III-D targeting 5-b resolution. In this case, we

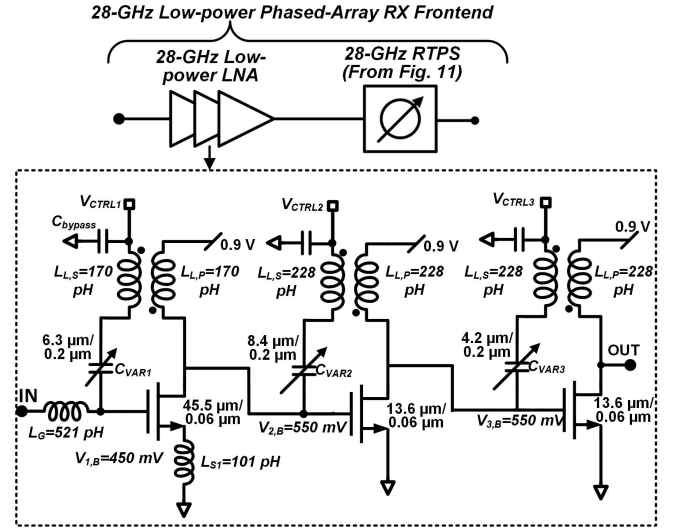


Fig. 14. Schematic of low-power RX phased-array front end, including three-stage low-power single-ended 28-GHz LNA with neutralization through transformer and varactor.

restrict the varactor control voltages, such that $V_{C1} = V_{C2}$ and $V_{C3} = V_{C4}$. Measured RTPS performance are compared with simulations in Section VI.

V. 28-GHz PHASED-ARRAY RX FRONT END

The RTPS is incorporated in a 28-GHz low-power phased-array RX front end with an LNA preceding the passive RTPS to provide sufficient gain to achieve overall low NF (Fig. 14). Given the ~ 8 -dB RTPS insertion loss, ~ 18 -dB LNA gain is targeted to achieve overall 10-dB gain in the front end. Single-stage amplifier gain at mm-wave in 65-nm CMOS is limited by the feedback capacitance between gate and drain, C_{GD} . A single-ended neutralization technique based on capacitive feedback using a transformer at the output has been presented in [32] to achieve higher gain for the same power consumption. This approach achieves higher gain [32] when compared with other neutralization-based techniques [33]–[35] while avoiding challenges related to inductor size for neutralization in a single-ended LNA. A wideband 60-GHz application ($\sim 15\%$) is targeted in [32], and hence, the transformer is used to provide wideband matching and to generate the neutralization signal. However, the 28-GHz band targeted in this paper is relatively narrowband (27.5–28.35 GHz). Therefore, we modify the topology in Fig. 15(a) to tap the output from the primary of the load transformer [Fig. 15(b)], which helps avoid losses due to nonideal coupling between primary and secondary. Implementing the feedback through a varactor allows compensation for modeling errors and intrinsic process variation. The schematic of the three-stage single-ended LNA with neutralization in each stage is shown in Fig. 14, where the source-degenerated first stage is biased for low noise and the following stages are biased to provide high gain. The varactors are implemented using the length of 200 nm and the finger width of 700 nm. The first-, second-, and third-stage varactor have 9, 12, and 6 fingers, respectively, providing $8.3 \text{ fF} \rightarrow 24.5 \text{ fF}$, $11.1 \text{ fF} \rightarrow 32.2 \text{ fF}$, and

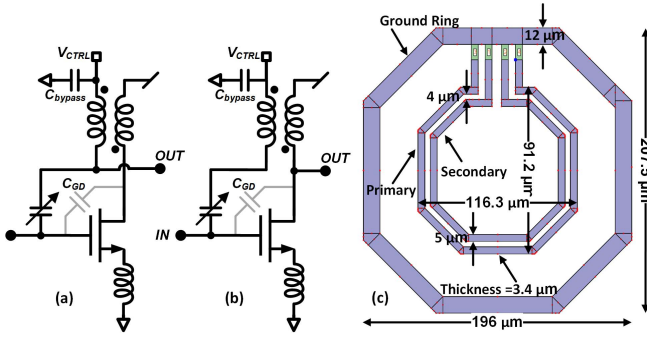


Fig. 15. (a), (b) Single-ended neutralization schemes to cancel C_{GD} for a single-ended amplifier using transformer at output to generate antiphase signal and feedback through varactor which provides variable feedback. (c) Layout of the proposed transformer used in the first stage of LNA. A similar topology is used in Stages 2 and 3.

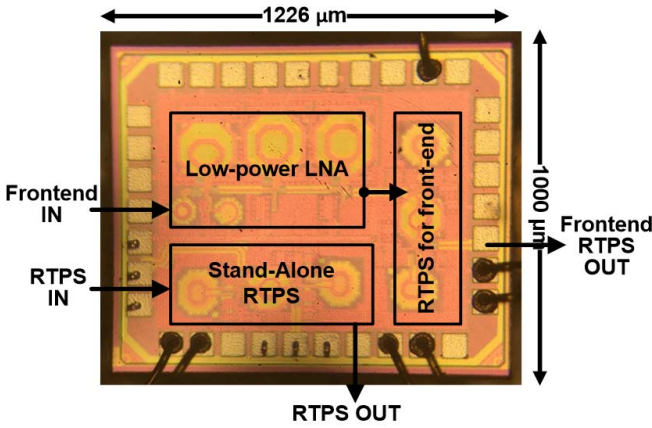


Fig. 16. Die photograph of implemented 28-GHz RX front-end and stand-alone RTPS.

5.6 fF $\rightarrow$ 16.7 fF of variable capacitance, respectively. The layout of the transformer in the first LNA stage is shown in Fig. 15(c). The transformer primary and secondary in Stages 2 and 3 are designed for higher primary and secondary inductances but follow the same topology. The transformers are designed in the top metal layer (the thickness of $3.4 \mu\text{m}$) using HyperLynx IE3D EM simulator. The three-stage LNA uses a common-source amplifier in each stage to minimize supply voltage (~ 0.9 V) and power consumption (~ 10 mW). Simulated performance is compared with measurement in Section VI.

VI. MEASUREMENT RESULTS

The phased-array RX front end (Fig. 16) was implemented in a 65-nm CMOS process with a $3.4\text{-}\mu\text{m}$ -thick top metal layer. The IC included a stand-alone test structure for characterizing the 28-GHz RTPS. The front-end and stand-alone RTPS s -parameters are measured using a probe-based setup with a Keysight 5227A network analyzer.

A. Measured 28-GHz RTPS Performance

Fig. 17 shows measured stand-alone 28-GHz RTPS $|S_{21}|$ and phase shift as Z_{T1} and Z_{T2} is varied by varying

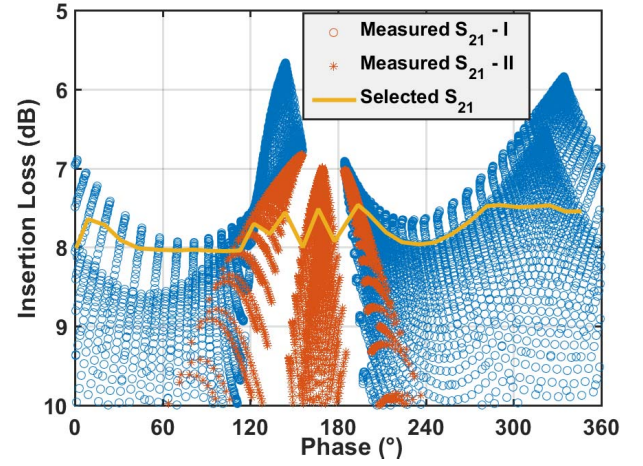


Fig. 17. Measured stand-alone RTPS insertion loss and phase shift across load settings at 28 GHz.

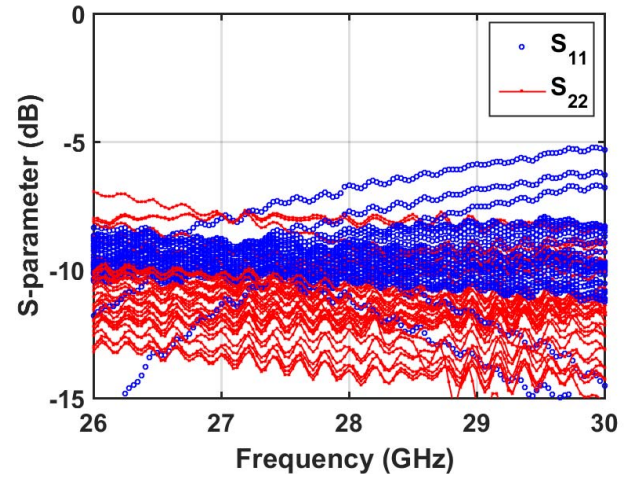


Fig. 18. Measured stand-alone RTPS input and output match across load settings from Fig. 17.

V_{C1} , V_{C2} , V_{C3} , and V_{C4} . The varactor control voltages are varied with 12.5-mV step size across a 0–2 V range (7.3 b). Measurements corresponding to $V_{C1} = V_{C3}$ and $V_{C2} = V_{C4}$ are in blue. The worse case loss in this occurs when C_1 , C_2 , C_3 , and C_4 are in the middle of their capacitance range as can be seen from simulated data in Fig. 9. Data corresponding to measurements where $V_{C1} = V_{C2}$ and $V_{C3} = V_{C4}$ are in red. The latter set of measurements are used to increase the phase shift range from 332° to 360° , as described in Section IV. In this case, insertion loss is degraded with higher asymmetry between coupled- and through-port terminations due to degradation in input and output match. However, in both cases, the high-loss settings are not used. As shown in Fig. 17, a subset of the varactor control voltage settings can be selected to achieve 7.75 ± 0.3 dB loss and full 360° phase shift with 5-b resolution. Fig. 18 shows the measured input and output match across frequency for all settings selected to achieve uniform loss across phase shift. The worse case return loss of 6.7 dB occurs for the settings when unequal terminations are applied to the coupler as described in Section III-D. This

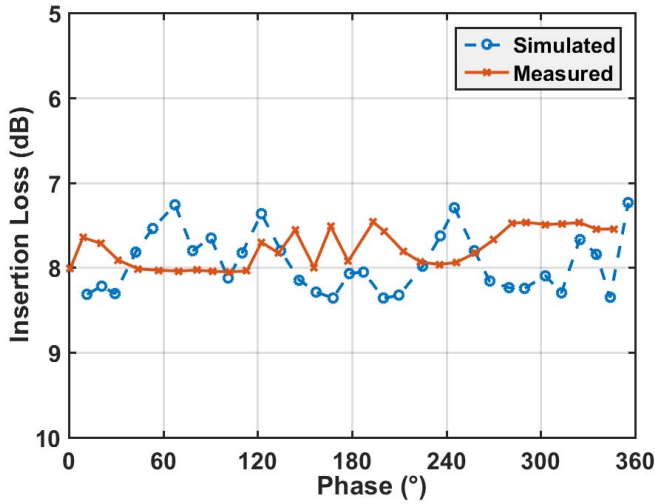


Fig. 19. Measured versus simulated performance for stand-alone RTPS at 28 GHz.

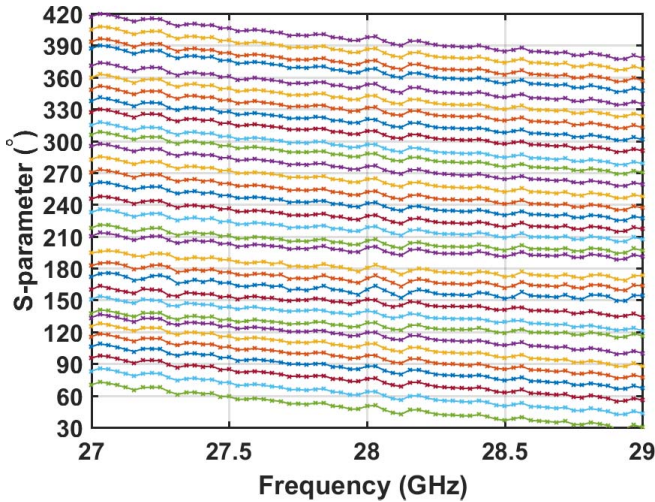


Fig. 20. Measured stand-alone RTPS phase shift across frequency.

is poorer than the simulated 10-dB return loss and is likely due to pad parasitics and modeling errors in the coupler. Fig. 19 compares the setting selection based on simulated and measured data, demonstrating good match across insertion loss and phase shift. The proposed approach utilizes independent control of four RTPS varactors as opposed to two in [7], necessitating four ~ 8 -b voltage DACs or two DACs with multiplexers to reconfigure outputs for the two voltage configurations described earlier. However, DAC power consumption can be low, since operation frequency must only be high enough to support phase shift update rates (tens of megahertz during beam direction calibration in an mm-wave link). For example, similar voltage DACs in 130-nm CMOS in [7] consume ~ 1.5 mW. Additionally, the larger phase shift achieved in the RTPS eliminates other phase shifters used in [7] and [26] reducing overall area and power consumption.

The measured phase shift across frequency for the stand-alone RTPS is shown in Fig. 20 for the subset of settings that correspond to uniform insertion loss in Fig. 17 for

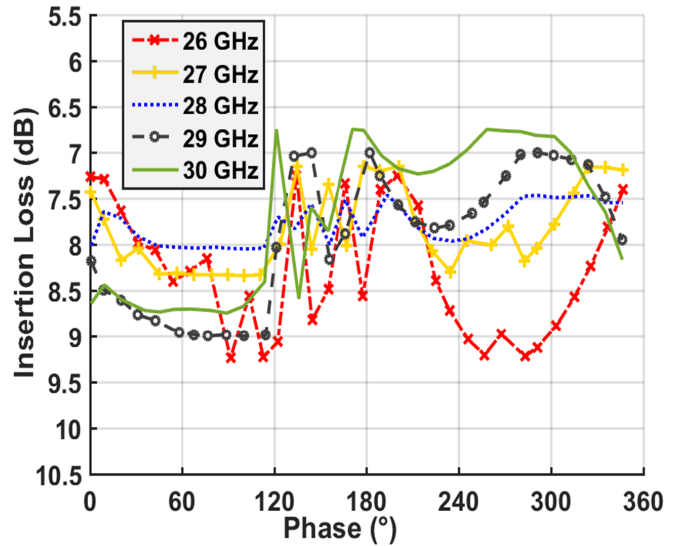


Fig. 21. Measured RTPS performance for different center frequencies. Phase shifter settings are optimized for each center frequency.

5-b phase resolution at 28 GHz. For these settings, an rms error is restricted to 0.3° with loss variation within ± 0.3 dB. In Fig. 20, the pi-network load in Fig. 6(a) in the proposed approach is optimized at single frequency. The narrowband termination design introduces frequency dependence and hence operation at other center frequencies requires recalculation of selected phase shift settings. Fig. 21 shows insertion loss and phase shift performance where optimal settings are recalculated at frequencies from 26 to 30 GHz. As shown in Fig. 21, recalculating phase shift settings can result in 8 ± 1.25 dB insertion loss from 26 to 30 GHz while achieving 5-b resolution and 360° range. However, larger loss variation can be observed at 26 and 30 GHz demonstrating limitations due to coupler and variable termination bandwidths.

The performance of the stand-alone RTPS is compared with the state of the art in Table I. The proposed RTPS achieves lower insertion loss for full 360° range (the loss is comparable with [7], which has 180° range). The insertion loss is comparable with the state-of-the-art STPS [20], and however, the proposed RTPS achieves higher resolution, smaller area, and better performance across frequency as discussed in Section II.

B. Measured 28-GHz RX Frontend Performance

In this section, we present the measured performance of the 28-GHz front end that includes the LNA and RTPS (Fig. 14). Two-port S -parameter measurements were performed across different varactor voltages. Control voltages for varactors in were swept in measurement to find optimal gain settings. Since the optimal capacitance depends upon C_{GD} , optimal settings will vary across process and temperature. Fig. 22 compares measured and simulated s -parameters in the lowest loss setting demonstrating good matching for S_{11} and S_{21} , with a shift in frequency for S_{22} . Given the ~ 5.5 -dB RTPS loss from Fig. 17, measurements indicate ~ 17.5 -dB gain for

TABLE I
STATE-OF-THE-ART mm-WAVE PHASE SHIFTERS

Approach[Reference]	Series RTPS [18]	Hybrid [38]	RTPS [26]	STPS [21]	This work
Process	180nm CMOS	65nm CMOS	130nm BiCMOS	65nm CMOS	65nm CMOS
Freq.(GHz)	24	60	94	28	28
Phase Range (°)	360°	360°	180°	360°	360°
Phase Resolution	Continuous	11.25°	11.25°	22.5°	11.25°
RMS Phase Error (°)	NA	4.4°	~2.5°	8.98°	0.3°
Insertion Loss (dB)	10.1 to 12.5	12.3 to 16.3	6.9 to 7.9	5.6 to 7.6	7.45 to 8.05
Max gain error (dB)	1.2	2	0.5	1	0.3
Return Loss (dB)	20	13	NA	12	6.7
Area (mm ²)	0.33	0.094	0.12	0.23	0.16

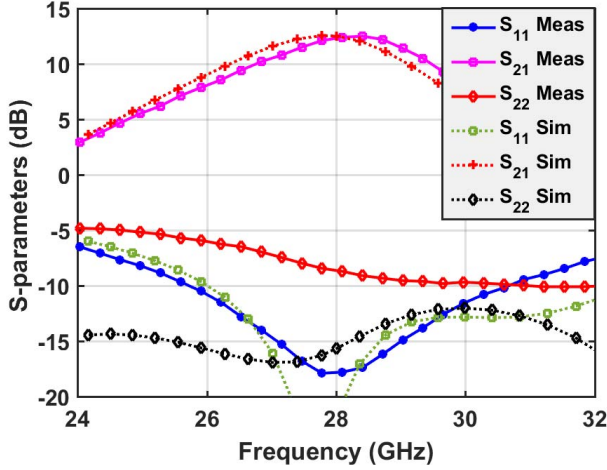


Fig. 22. Comparison of measured and simulated S -parameters for the RX front end.

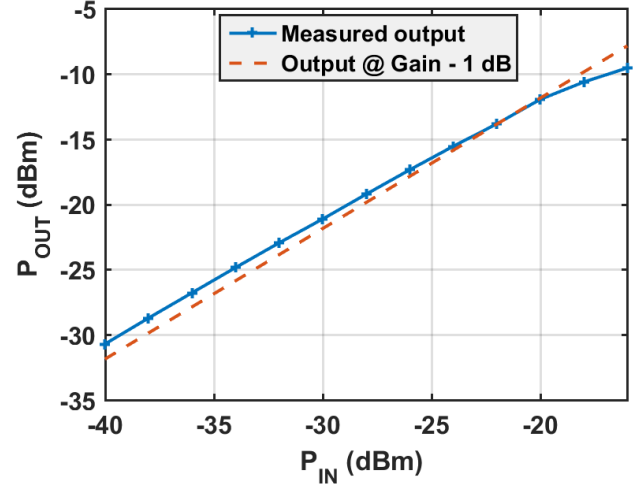


Fig. 24. Measured RX front-end gain compression at 28 GHz.

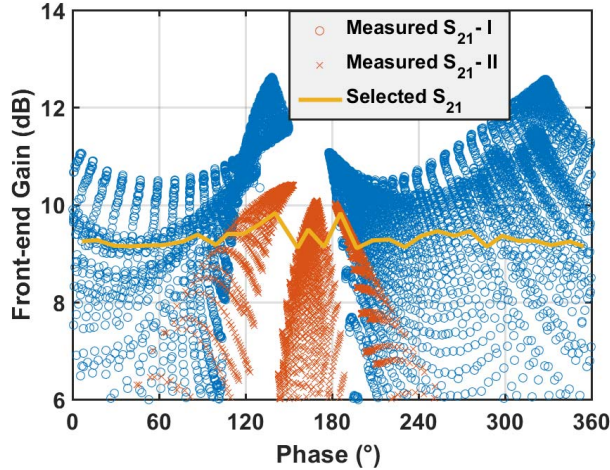


Fig. 23. Measurements RX front-end gain and phase shift at 28 GHz across varactor settings.

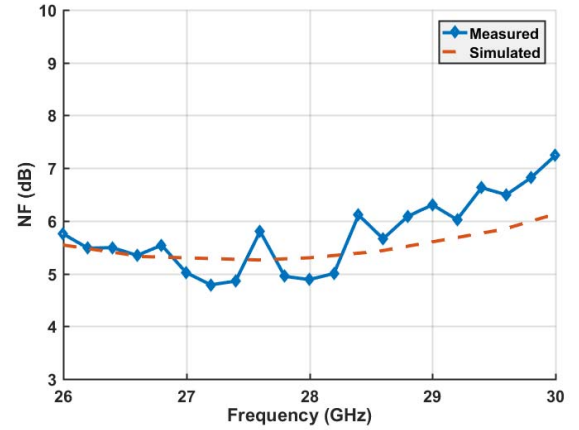


Fig. 25. Measured and simulated RX front-end NF.

the three-stage LNA, which is within 1 dB of simulated performance.

Measured front-end gain and phase across RTPS control voltages, similar to that shown for stand-alone RTPS in Fig. 17, are shown in Fig. 23. Measured performance shows overall front-end gain of 9.5 dB with gain variation of ± 0.4 dB. The front end achieves 360° phase shift range with

11.25° resolution and 0.4° rms phase error at 28 GHz. The measured input-referred 1-dB compression point of the RX front end is approximately -22 dBm at 28 GHz (Fig. 24). Two-tone tests with 28- and 28.02-GHz input indicate an input-referred IP3 of -11.5 dBm. Measured front-end NF is compared with simulated in Fig. 25(b), demonstrating better than 5.5-dB NF at 28 GHz. The low-power RX front end consumes 10-mW power from a 0.9-V supply. The front end is compared with the state of the art in Table II. The implemented

TABLE II
STATE-OF-THE-ART mm-WAVE FRONT END

[Reference]	[37]	[38]	[39]	[13]	[40]	This work
Process	120nm SiGe	65nm CMOS	90nm CMOS	120nm SiGe	45nm CMOS SOI	65nm CMOS
Freq.(GHz)	60	61	60	34	27	28
Phase Range (°)	360°	360°	360°	360°	360°	360°
Phase Resolution	45°	22.5°	Continuous	22.5°	11.25°	11.25°
RMS Phase Error (°)	NA	7°	NA	<4°	4°	0.4°
Gain (dB)	14	12	11.3	11	12.2	9.5
Gain variation (dB)	±1.2	±1.7	±1.85	±1.5	±1	±0.4
PI dB (dBm)	-33.5	-16	-20	-28	-8	-22
NF (dB)	<6.9	<7.2	6.3	<3.4	4	<5.5
Power consumption (mW)	66	<78	18	11	42	10
Area (mm ²)	4.63	1.6	0.89	0.3	1.75	0.32

front end achieves low power consumption and full 360° phase shift range with low rms phase error and loss variation.

VII. CONCLUSION

RF-path phase shifting promises low power consumption and compact area for large-scale arrays. This paper addresses the challenge of generating variable phase shift in the signal path at mm-wave, demonstrating a 28-GHz low-power phased-array RX front end in 65-nm CMOS for RF-path phase-shifting phased arrays. The front end incorporates a state-of-the-art RTPS that achieves full 360° phase shift with 5-b phase resolution at 28 GHz, with <1° rms phase error. A systematic approach to determine RTPS load-network parameters is presented that enables low insertion loss, uniform loss across shift, and large phase shift range simultaneously. Single-ended neutralization network in the LNA ensures high gain without degrading power consumption. The ~10-mW power consumption, 9.5-dB gain, and <5.5 dB NF demonstrate the suitability of the phased-array RX front end for large-scale arrays targeting 5G wireless links.

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RF and millimeter-wave

Robin Garg (M'16) received the B.Tech. degree in electrical engineering from the IIT Madras, Chennai, India, in 2010, and the M.S. degree in electrical engineering from the Oregon State University, Corvallis, OR, USA, in 2016.

From 2010 to 2014, he was an RF Design Engineer with the Wireless Connectivity Solutions Group, Texas Instruments, Bangalore, India. He is currently a Senior Design Engineer with Cypress Semiconductor Corporation, Lynnwood, WA, USA. His current research interests include wireless communications, integrated circuits, and systems design.



Arun S. Natarajan (M'07) received the B.Tech. degree from IIT Madras, Chennai, India, in 2001, and the M.S. and Ph.D. degrees from the California Institute of Technology, Pasadena, CA, USA, in 2003 and 2007, respectively, all in electrical engineering.

From 2007 to 2012, he was a Research Staff Member with the IBM T. J. Watson Research Center, Yorktown Heights, NY, USA, where he was involved in millimeter-wave (mm-wave) phased arrays for multigigabit-per-second data links and airborne radar and on self-healing circuits for increased yield in submicrometer process technologies. In 2012, he joined the School of Electrical Engineering and Computer Science, Oregon State University, Corvallis, OR, USA, as an Assistant Professor. His current research interests include RF, mm-wave and sub-mm-wave integrated circuits and systems for high-speed wireless communication and imaging.

Dr. Natarajan was a recipient of the National Talent Search Scholarship from the Government of India from 1995 to 2000, the Caltech Atwood Fellowship in 2001, the IBM Research Fellowship in 2005, the 2011 Pat Goldberg Memorial Award for the best paper in computer science, electrical engineering, and mathematics published by IBM Research, the CDADIC Best Faculty Project Award in 2014 and 2016, the NSF CAREER Award in 2016, the Oregon State University Engelbrecht Young Faculty Award in 2016, and the DARPA Young Faculty Award in 2017. He has served on the Technical Program Committee of the IEEE International Solid-State Circuits Conference, the IEEE Radio-Frequency Integrated Circuits Conference, and the IEEE Compound Semiconductors IC Symposium. He is also an Associate Editor of the IEEE TRANSACTIONS ON VLSI SYSTEMS and the IEEE TRANSACTIONS ON MICROWAVE THEORY AND TECHNIQUES.